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Astrostatistical inference for extremes of solar spectral irradiance in the SORCE/XPS record

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Introduction

Space mission archives are our main source for studying how extreme solar variability can be. However, an extreme estimated from such an archive is not purely a property of the Sun. It also reflects which days the instrument observed, how each daily value was produced, and how gaps were handled. We study this problem for daily changes in soft X-ray irradiance measured by SORCE/XPS, using Generalized Extreme Value (GEV) models [3].

Data

Level 3 daily mean irradiance from photodiode 7 of SORCE/XPS (nominal band 0.1 to 7.0 nm), from 2005 to 2019 [1, 2]. Of 5478 calendar days, 5073 were observed (92.6%). The record has 42 gaps.

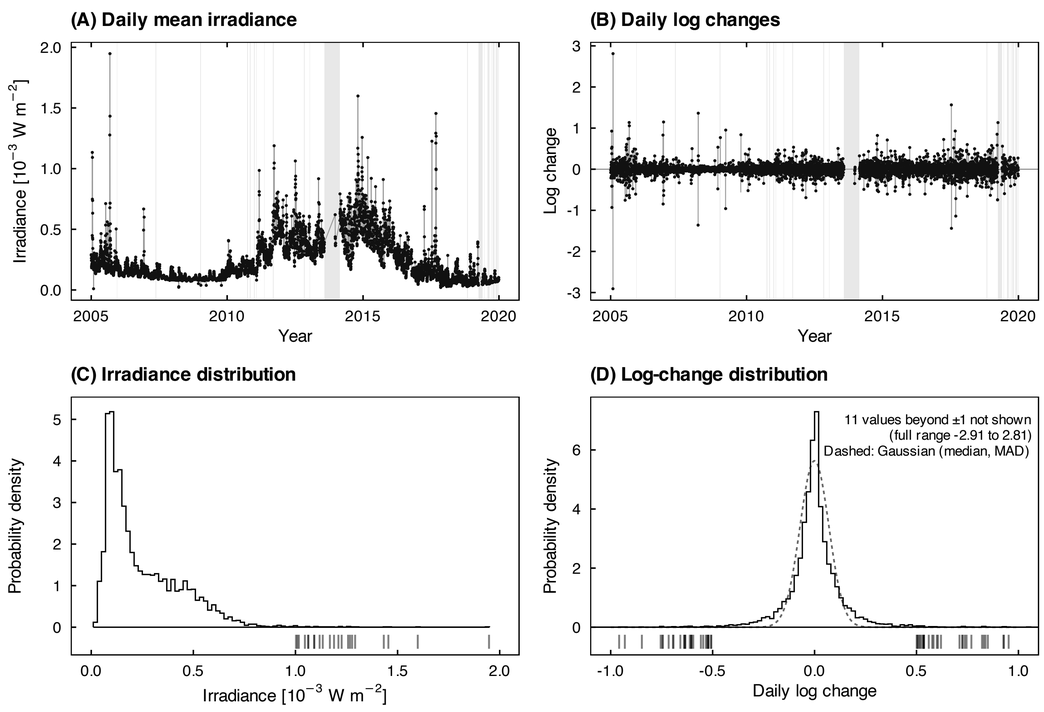


Figure 1. (A) Daily irradiance and (B) daily log changes, 2005 to 2019; (C, D) their distributions. Grey bands: gaps of two or more days.

Methods

- Daily logarithmic change $r_t = \log(X_t / X_{t-1})$, computed only between consecutive observed days. Gaps are not filled.
- Increases: **maxima** of 60 day blocks (85 blocks).
- Decreases: sign reversed **minima** of 15 day blocks (341 blocks).
- GEV fitted by maximum likelihood, with 95% intervals from a parametric bootstrap (5000 samples), and annualized return levels.

The **GEV density** [4] is

$$f_{\mathrm{GEV}}(y;\mu,\sigma,\xi) = \begin{cases} \frac{1}{\sigma}\left[1+\xi\left(\frac{y-\mu}{\sigma}\right)\right]^{-1/\xi-1}\exp\left\{-\left[1+\xi\left(\frac{y-\mu}{\sigma}\right)\right]^{-1/\xi}\right\}, & \xi \neq 0, \\ \frac{1}{\sigma}\exp\left\{-\left(\frac{y-\mu}{\sigma}\right)-\exp\left[-\left(\frac{y-\mu}{\sigma}\right)\right]\right\}, & \xi = 0. \end{cases}$$

With location $\mu \in \mathbb{R}$, scale $\sigma > 0$ and shape $\xi \in \mathbb{R}$.
Its support for $\xi \neq 0$ is $1 + \xi\,(y-\mu)/\sigma > 0$.

GEV estimates

Both tails have positive shape estimates, and both bootstrap intervals exclude zero. This is compatible with Fréchet type (heavy) tails under this construction. For increases the interval nearly reaches zero, and removing the largest block (summer 2017) lowers the estimate by 17%.

Table 1. GEV estimates by maximum likelihood, 95% bootstrap intervals.

	q (days)	k	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\xi}$
Increases (block maxima)	60	85	0.276 [0.236, 0.320]	0.173 [0.138, 0.205]	0.225 [0.032, 0.418]
Decreases (block minima)	15	341	0.121 [0.110, 0.133]	0.091 [0.081, 0.100]	0.345 [0.251, 0.441]

Return levels

For blocks of q days, the T year return level $z_{T,q}$ is exceeded on average once every T years, $\mathbb{P}(Y > z_{T,q}) = q/(365.25\,T)$, where

$$z_{T,q} = \mu + \frac{\sigma}{\xi}\left[\left\{-\log\left(1-\frac{q}{365.25\,T}\right)\right\}^{-\xi} - 1\right].$$

Table 2. Return levels $\hat{z}$ (log change scale), 95% bootstrap intervals.

	Increases (block **maxima**)		Decreases (block **minima**)	
T (years)	$\hat{z}^{+}_{T,60}$	Increase % [95% CI]	$\hat{z}^{-}_{T,15}$	Decrease % [95% CI]
2	0.842	132.2 [96.1, 184.4]	0.861	57.7 [50.1, 65.8]
5	1.158	218.3 [135.3, 388.1]	1.237	71.0 [60.9, 80.6]
20	1.769	486.5 [207.8, 1779.3]	2.086	87.6 [76.0, 95.5]
50	2.289	886.4 [266.7, 7536.5]	2.917	94.6 [84.5, 99.1]
100	2.760	1479.3 [321.5, 32457.5]	3.744	97.6 [89.7, 99.8]

Percentages are relative to the previous day; decreases are bounded at 100%.

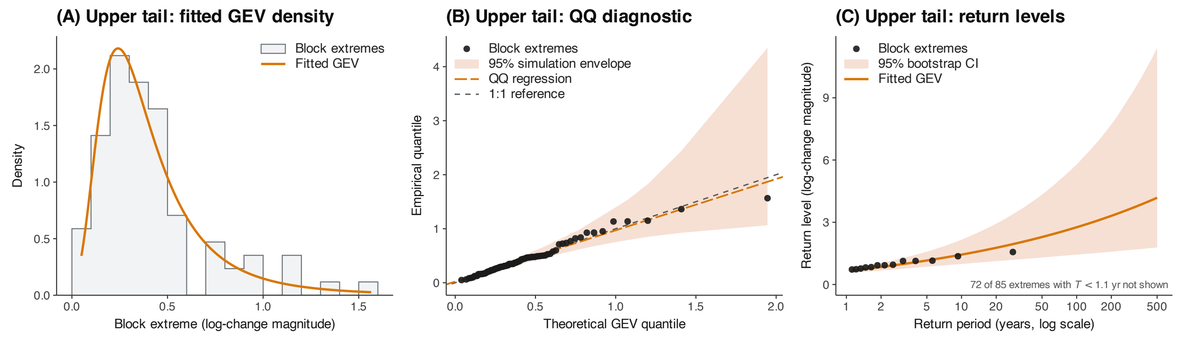


Figure 2. Increases (60 day block maxima): fitted GEV density, quantile plot with 95% envelope, and return levels with 95% bootstrap band.

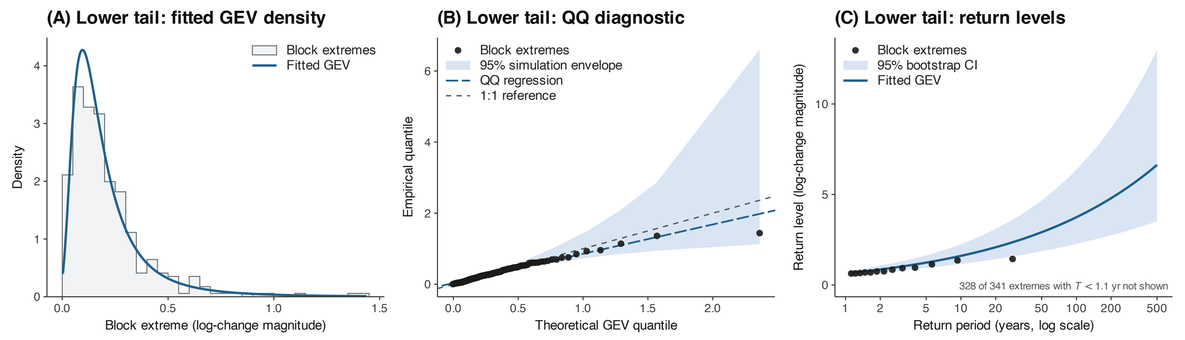


Figure 3. Decreases (15 day block minima), same panels as Figure 2.

Sensitivity

- Interpolating across every gap lowers both shape estimates by about 25%, so it makes the tails lighter.
- A solar activity covariate (F10.7) lowered AIC in both tails, but the implied shift in return levels changes direction across tails and models.
- Photodiode 1, with the same nominal band, also gives positive shape estimates. This is concordance, not independent confirmation, since both detectors share sampling and processing.

Conclusions

Tail estimates from mission data describe the processed record, not the latent solar irradiance. They are meaningful only alongside their sensitivity to how observations become an analytical series. Even for a complex record, with gaps, missing days and systematic errors, extreme value analysis remains essential, and the study of extremes deserves a larger place in astronomy.

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References

[1] Woods, T. N. (2020). SORCE XPS Level 3 Solar Spectral Irradiance Daily Means V012 [Dataset]. GES DISC. doi:10.5067/ZBKF34FGDCLT
[2] Woods, T. N. and Elliott, J. (2022). Solar radiation and climate experiment (SORCE) X-ray photometer system (XPS): final data-processing algorithms. *Solar Physics* 297. doi:10.1007/s11207-022-01997-4
[3] Kotz, S. and Nadarajah, S. (2000). *Extreme Value Distributions: Theory and Applications* (Singapore: World Scientific)
[4] von Mises, R. (1936). La distribution de la plus grande de n valeurs. *Revue Mathématique de l'Union Interbalkanique* 1, 141–160

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